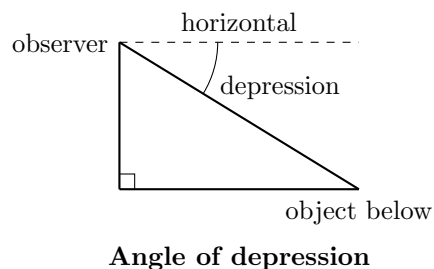
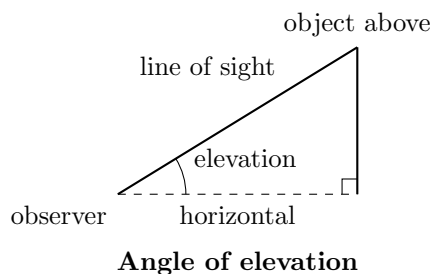


Right Triangles in Elevation and Depression Problems

High School Geometry

angles of elevation and depression: finding a side, finding an angle, and two-triangle setups

How the two angles are measured. Both are measured from the *horizontal*, never from the vertical. The reference diagram below shows the labelling every problem on this sheet uses. *No values are shown — use the numbers given in each problem.*



Draw and label the right triangle before you compute. State the ratio you are using, and round each answer as the problem asks. Keep your calculator in degree mode.

1. A surveyor stands on level ground 40 ft from the base of a tree. The angle of elevation from the ground at her feet to the top of the tree is 32° . How tall is the tree, in feet, to the nearest hundredth of a foot?

Answer: _____ ft

2. From the top of a lighthouse 85 m above sea level, the angle of depression to a boat is 12° . How far is the boat from the foot of the lighthouse, measured along the water, to the nearest tenth of a metre?

Answer: _____ m

3. A wheelchair ramp rises 3 ft over a horizontal run of 20 ft. Find the angle of elevation of the ramp, in degrees, to the nearest tenth of a degree.

Answer: _____ degrees

4. A surveyor's eye is 45 m horizontally from the base of a tower, and the top of the tower is 24 m above her eye level. Let θ be the angle of elevation from her eye to the top of the tower. Write $\tan \theta$ as a fraction in lowest terms. (Do not find θ .)

Answer: _____

5. A kite is flying at the end of a taut string 150 ft long. The string makes an angle of elevation of 28° with the ground. How high above the ground is the kite, in feet, to the nearest hundredth of a foot? (Assume the string is held at ground level.)

Answer: _____ ft

6. A plane is cruising at an altitude of 3200 ft. The pilot measures the angle of depression to an airport ahead as 15° . How far is the airport from the point directly beneath the plane, measured along the ground, to the nearest foot?

Answer: _____ ft

7. A student whose eye is 1.6 m above the ground stands 12 m from the foot of a flagpole on level ground. The angle of elevation from her eye to the top of the flagpole is 38° . Find the height of the flagpole above the ground, in metres, to the nearest hundredth of a metre. (The extra measurement is not decoration — decide where it belongs.)

Answer: _____ m

8. A ranger stands at the edge of a cliff 50 m above the sea and looks out at two boats directly in line with the cliff. The angle of depression to the nearer boat is 35° and to the farther boat is 20° . How far apart are the two boats, in metres, to the nearest hundredth of a metre?

Answer: _____ m